

Multilabel Movie Genre Classification from Poster Images Using Deep-Learning Frameworks

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Abstract—Multilabel genre classification of movie posters is a challenging computer vision problem where a single image may simultaneously convey multiple genre signals. This paper adapts and extends the framework proposed by Unal et al. (2023), applying five state of the art pretrained convolutional neural network architectures, namely VGG16, ResNet50, DenseNet121, MobileNet, and ConvNeXt, to a dataset of 300 IMDB movie posters annotated with 14 genre labels. To mitigate overfitting on this small dataset, each backbone is adapted with a simplified linear classification head with dropout, and trained using Binary Cross-Entropy with Logits Loss with positive class weighting to address genre imbalance. A decision threshold of $\tau = 0.5$ is applied alongside a Top 4 genre selection strategy, with Hamming Loss as the primary evaluation metric. Experimental results show that ResNet50 achieves the lowest Hamming Loss in the frozen phase, outperforming VGG16, DenseNet121, MobileNet and ConvNeXt. Full backbone fine tuning of the top 2 models reveals that the frozen phase generalizes better on this small dataset of 250 training images. These findings are consistent with prior literature and validate the applicability of large pretrained models to small scale multilabel poster classification tasks.

Index Terms—multilabel classification, movie genre, convolutional neural network, transfer learning, Hamming loss, image classification

I. INTRODUCTION

Movie posters serve as the primary visual interface between a film and its potential audience. Studios invest considerable effort in designing posters that communicate genre, mood, and narrative through visual cues such as color palette, character pose, typography, and compositional elements [2]. Automatically inferring genre from poster imagery is therefore both commercially relevant, enabling recommendation systems and content tagging pipelines, and scientifically interesting as a challenging multilabel visual recognition task.

Multilabel classification assigns one or more labels to each instance, contrasting with multiclass problems where labels are mutually exclusive. In the context of movie genre prediction, a single film such as *Loki* may belong to Drama, Fantasy, Animation, Comedy, Family, and Action simultaneously, making the problem inherently multilabel [1]. Challenges include label imbalance, where Drama appears far more frequently than Musical, and the difficulty of choosing appropriate evaluation metrics, since element wise accuracy fails to capture partial correctness.

Transfer learning from large ImageNet pretrained models has emerged as the dominant paradigm for vision tasks with limited labeled data [4]. Unal et al. [1] compared five such architectures, namely VGG16, ResNet50, DenseNet, MobileNet, and ConvNeXt, on a large IMDB poster dataset and reported that DenseNet and ConvNeXt achieved the highest accuracy above 90%. This work applies their methodology to a smaller dataset of 300 annotated movie posters, investigating whether the same ranking of architectures holds under a more constrained data regime with 14 genre categories.

The main contributions of this paper are the development of a complete multilabel poster genre classification pipeline that replicates the approach proposed by Unal et al. [1] using five different deep learning architectures. In addition, this study provides a systematic evaluation employing multiple performance metrics, including Hamming Loss, F1-score, and Precision, under both threshold-based ($\tau = 0.5$) and Top-4 prediction strategies. Furthermore, the paper analyzes the performance differences between frozen-backbone transfer learning and full fine-tuning approaches on a limited training dataset consisting of 250 images.

To support reproducibility of our experiments, the complete PyTorch training pipeline, dataset splits, and final results are publicly available in our GitHub repository: <https://github.com/DevanWestley/ComVis>.

II. RELATED WORK

Genre classification from visual poster content has been explored through both traditional and deep learning approaches. Chu et al. [2] used a modified AlexNet combined with YOLO object detection for 23-genre multilabel classification on the IMDB dataset, achieving a vector accuracy of 18.73%. Sung et al. [3] compared ResNet-50, VGG-16, and DenseNet-169 on a 7-genre Kaggle dataset, with DenseNet-169 obtaining an F1-score of 0.77. Kundalia et al. [5] applied transfer-learned InceptionV3 to 12-genre poster classification and reported 84.82% accuracy.

More recently, Unal et al. [1] conducted the most comprehensive comparison of modern architectures on this task, introducing ConvNeXt [6], a ResNet modernized to match Vision Transformer performance while retaining a fully convolutional structure, as a novel backbone. Their dataset of 29,432 IMDB

posters across 20 genres, split using iterative stratification [8] to preserve label distribution, serves as the reference for this work.

III. METHODOLOGY

A. Dataset

The dataset consists of 300 movie poster images with filename labels, paired by multilabel genre annotations with 14 binary genre columns. Genre values are encoded using multi-hot encoding where $y_k = 1$ if the film belongs to genre k and 0 otherwise:

$$\mathbf{y} = [y_1, y_2, \dots, y_{14}] \in \{0, 1\}^{14} \quad (1)$$

The dataset is partitioned into 250 training images and 50 test images. As illustrated in Fig. 1, the genre distribution is significantly imbalanced where Action and Drama are the most frequent categories while Musical has fewer than 10 representatives, consistent with the pattern observed by Unal et al. [1] in the full IMDB dataset.

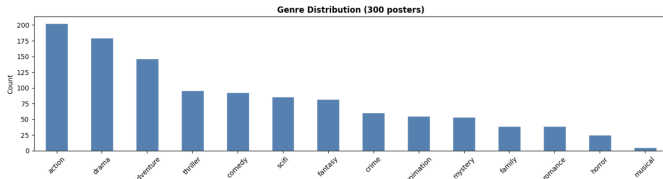


Fig. 1. Distribution of the number of movies by genre.

B. Data Preprocessing and Augmentation

All images are resized to 256×256 pixels, consistent with the input specification of Unal et al. [1]. ImageNet mean and standard deviation are applied for normalization:

$$\hat{x} = \frac{x - \mu_{\text{IN}}}{\sigma_{\text{IN}}} \quad (2)$$

where $\mu_{\text{IN}} = [0.485, 0.456, 0.406]$ and $\sigma_{\text{IN}} = [0.229, 0.224, 0.225]$.

To combat overfitting on the small 250-image training set, the following stochastic augmentation pipeline is applied exclusively during training: (1) random horizontal flip with $p = 0.5$, (2) random rotation in $[-10^\circ, +10^\circ]$, and (3) color jitter with brightness ± 0.3 , contrast ± 0.3 , saturation ± 0.2 , and hue ± 0.1 . Fig. 2 illustrates four augmented variants of a representative training poster alongside the original, demonstrating the diversity introduced by this pipeline.



Fig. 2. Data augmentation examples.

C. Model Architecture

Following Unal et al. [1], five pretrained CNN backbones are employed: VGG16 [9], ResNet50 [10], DenseNet121 [12], MobileNetV2 [13], and ConvNeXt-Tiny [6] with general architecture model illustrated in Fig. 3.

To prevent severe overfitting on our highly constrained 250-image training set, the massive multi-layer perceptron head used in the original paper is replaced with a simplified linear probe. The original classification head of each backbone is removed, and the extracted features h_{backbone} are passed through a 1D Batch Normalization (BN) layer, a Dropout layer ($p = 0.5$) for regularization, and a single Linear projection layer. The forward pass for the new classification head is thus simplified to:

$$\hat{\mathbf{y}} = \mathbf{W} \cdot \text{Dropout}(\text{BN}(h_{\text{backbone}})) + \mathbf{b} \quad (3)$$

where $\mathbf{W} \in \mathbb{R}^{K \times d_{\text{backbone}}}$ and $\mathbf{b} \in \mathbb{R}^K$. The output logits $\hat{\mathbf{y}}$ are raw, with the Sigmoid activation applied only at inference to maintain numerical stability with the BCEWithLogitsLoss during training.

K=14 Multilabel Genre Classification Architecture (Phase 1: Frozen Backbone)

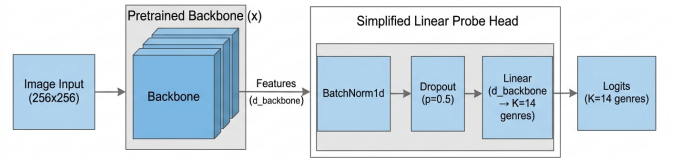


Fig. 3. General Architecture of the Model

D. Training Protocol

In Phase 1, all pretrained backbone weights are frozen, and only the simplified linear classification head is trained. This approach follows the standard transfer learning paradigm, allowing the model to leverage robust pretrained visual features while preventing severe overfitting on the small training set. In Phase 2, the top two models from Phase 1, selected based on the lowest Hamming Loss, undergo full backbone fine-tuning using a reduced learning rate. This second stage was designed to investigate whether unfreezing the pretrained features would allow them to adapt more effectively to the specific characteristics of the poster genre domain, or if the extreme data scarcity would instead lead to catastrophic forgetting.

E. Loss Function and Class Imbalance

BCEWithLogitsLoss is applied as the objective function, treating each of the $K = 14$ genre outputs as an independent

binary classification:

$$\mathcal{L} = -\frac{1}{N \cdot K} \sum_{i=1}^N \sum_{k=1}^K [y_{ik} \log \sigma(\hat{y}_{ik}) + (1 - y_{ik}) \log(1 - \sigma(\hat{y}_{ik}))] \quad (4)$$

To address the severe label imbalance shown in Fig. 1, per-genre positive class weights are computed from the training set as:

$$w_k = \frac{N - n_k^+}{n_k^+}, \quad k = 1, \dots, K \quad (5)$$

where n_k^+ is the number of training examples with $y_k = 1$. These weights penalize missed detections of rare genres proportionally.

F. Optimization

Adam [14] is used as the optimizer with initial learning rate $\eta = 10^{-3}$ (Phase 1) and $\eta = 10^{-5}$ (Phase 2), weight decay $\lambda = 10^{-4}$, and mini-batch size 32 for 50 epochs per phase. A multiplicative decay factor of $\gamma = 0.5$ is applied to systematically reduce the learning rate, ensuring robust convergence without relying on a separate validation partition.

G. Evaluation Metrics

Model performance is assessed using four metrics. Accuracy is the exact-match ratio over all test samples. Precision is the proportion of true positive predictions among all positive predictions. The F-score is the harmonic mean of precision and recall:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (6)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad F\text{-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

The primary metric is Hamming Loss [1], defined as the fraction of incorrectly predicted labels across all samples and genres:

$$\text{Hamming Loss} = \frac{\text{Number of incorrectly predicted labels}}{\text{Total number of labels}} \quad (8)$$

A lower Hamming Loss indicates better per-label prediction accuracy. Genre prediction employs two complementary strategies: a threshold strategy ($\tau = 0.5$, predicts all genres with Sigmoid probability ≥ 0.5) and a Top-4 strategy (selects the 4 genres with the highest probabilities regardless of their absolute values).

IV. EXPERIMENTAL RESULTS

All experiments were executed on Google Colab with a Tesla T4 GPU using PyTorch 2.x and Torchvision. Pretrained weights were obtained from the Torchvision model zoo and initialized from ImageNet.

TABLE I
PHASE 1 TRAINING RESULTS

Arch	Epoch	Train Loss	Val Loss	HL
VGG16	1	1.2253	1.0330	0.2343
	25	0.0564	1.9945	0.2171
	50	0.0344	2.3089	0.2043
ResNet50	1	1.0716	0.8729	0.2886
	25	0.3396	0.8665	0.2186
	50	0.2470	0.9476	0.1986
DenseNet121	1	1.0241	0.8771	0.3771
	25	0.4506	0.8103	0.2229
	50	0.3702	0.9128	0.2143
MobileNet	1	1.0484	0.8768	0.4000
	25	0.4440	0.8208	0.2386
	50	0.3666	0.9058	0.2100
ConvNeXt	1	1.0769	0.8618	0.3986
	25	0.4575	0.6973	0.2414
	50	0.3698	0.7381	0.2071

A. Training Curves

Fig. 4 shows the training and validation BCE loss (left axis, solid lines) and Hamming Loss (right axis, dashed lines) for all five architectures across training phase 1. In Phase 1, while the training BCE loss exhibits consistent monotonic reduction, the validation BCE loss diverges and increases after the initial epochs. This behavior indicates that the models begin to overfit on the small training set, making overconfident incorrect predictions that penalize the BCE objective. However, the validation Hamming Loss stabilizes or continues to improve, which justifies our custom early-stopping strategy of saving the model weights at the epoch with the lowest validation Hamming Loss.

Phase 2 fine-tuning of the top-2 models shows that unfreezing all layers with a learning rate of $\eta = 10^{-5}$ increases the Hamming Loss compared to the frozen phase. This confirms that full fine-tuning on a highly constrained dataset of only 250 training images causes the backbone features to overfit to the small training distribution. Table I presents the per-epoch training log for the architectures during Phase 1.

B. Test Set Evaluation

Table II presents the complete test-set metrics for all five frozen-backbone models and the two fine-tuned variants. ResNet50 achieves the best overall performance, recording the lowest Hamming Loss at $\tau = 0.5$ (0.1929), the lowest Top-4 Hamming Loss (0.1757), and the highest F1-Score (0.6423). However, VGG16 follows very closely behind (Hamming Loss of 0.2000) and actually outperforms ResNet50 in Exact Match Accuracy (0.1200 compared to 0.0400) while maintaining a highly competitive Precision (0.6491). This suggests that VGG16 adopts a more conservative prediction strategy, yielding highly accurate exact matches when it is confident, whereas ResNet50 finds a better global balance between Precision and Recall across all classes.

Finally, consistent with the well-documented behavior of large pretrained models on extremely limited data, full fine-

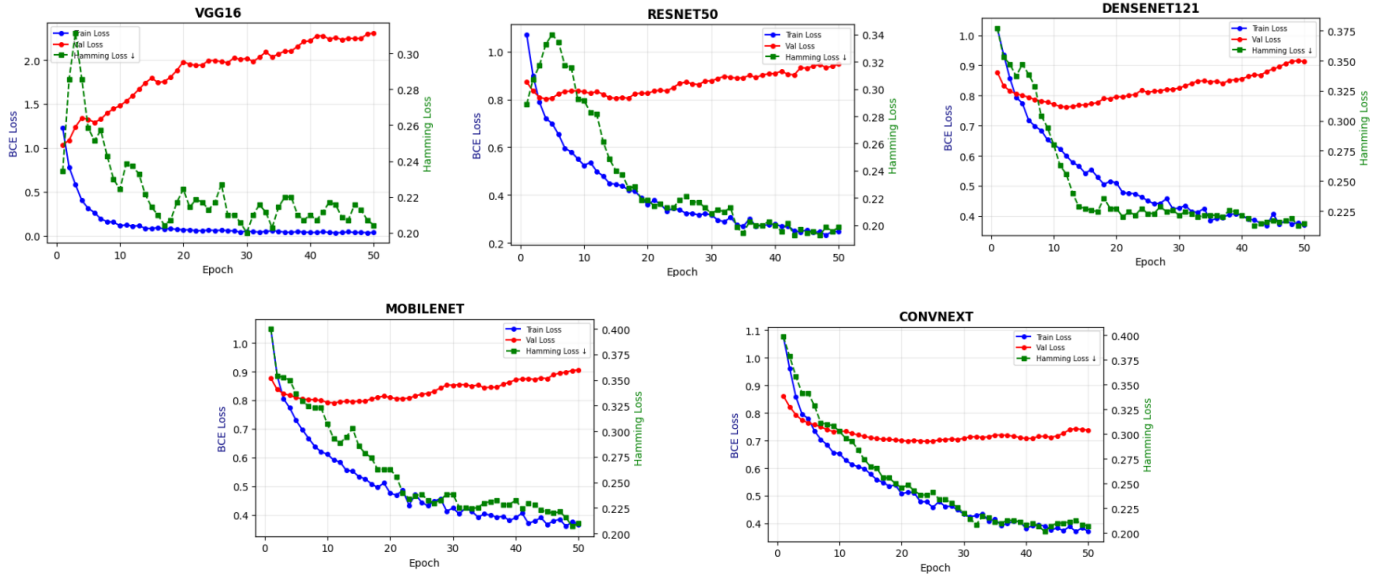


Fig. 4. Training curves for all five architectures during Phase 1.

TABLE II
FULL TEST SET EVALUATION RESULTS ($\tau = 0.5$, TOP-4).

Model	Phase	HL ($\tau = 0.5$) ↓	HL (TOP-4) ↓	Accuracy ↑	Precision ↑	F1-Score ↑
ResNet50	Frozen	0.1929	0.1757	0.0400	0.6456	0.6423
VGG16	Frozen	0.2000	0.2043	0.1200	0.6491	0.6084
ConvNeXt	Frozen	0.2014	0.1900	0.0200	0.6436	0.6369
MobileNet	Frozen	0.2071	0.1929	0.0400	0.6512	0.6229
DenseNet121	Frozen	0.2129	0.2043	0.1000	0.6232	0.6183
ResNet50 (FT)	Fine-tuned	0.3843	0.3271	0.0000	0.3804	0.4550
VGG16 (FT)	Fine-tuned	0.3429	0.2700	0.0000	0.4279	0.5204

tuning severely degrades performance for both top models. For instance, the Hamming Loss of ResNet50 nearly doubles from 0.1929 to 0.3843, empirically proving the vulnerability of these deep architectures under data scarcity.

C. Qualitative Prediction Example

Fig. 5 shows an example prediction for a test poster using the best ResNet50 frozen model. The model correctly identifies Romance, Crime, Drama, and Horror as the top-4 genres, all exceeding the threshold $\tau = 0.5$. Lower-probability genres such as Thriller and Musical are also predicted. Genres unlikely for this poster (Animation, Adventure, Family, Fantasy) receive near-zero probabilities, demonstrating that the model learns to suppress irrelevant genre activations.

V. DISCUSSION

The results demonstrate three key findings. First, ResNet50 generalizes best on this highly constrained 250-image training set during the frozen phase, achieving a Hamming Loss of 0.1929. This contrasts with the findings of Unal et al. [1], who reported ConvNeXt as the top performer on the large

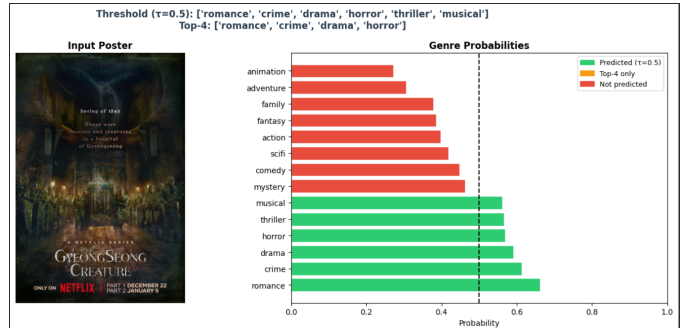


Fig. 5. Genre prediction test using the best ResNet50 model.

IMDb dataset. Our results indicate that older, highly robust architectures like ResNet50 and VGG16 adapt better to simple linear probing when data is scarce. Highly parameterized modern architectures like ConvNeXt may require significantly more data to effectively align their features to a new domain without overfitting.

Second, full fine-tuning degrades performance for both

top models on this dataset. The Hamming Loss increases significantly from 0.1929 to 0.3843 for ResNet50 after fine-tuning with $\eta = 10^{-5}$. This is expected because, with only 250 training images across 14 genre classes, the gradient updates from fine-tuning distort the well-calibrated ImageNet features faster than the small dataset can provide a corrective signal. This confirms that when labeled data is scarce, the frozen-backbone approach provides a much stronger inductive bias.

Third, the Top-4 Hamming Loss is consistently lower than the threshold-based Hamming Loss for all models (e.g., ResNet50 drops from 0.1929 to 0.1757), indicating that the model produces meaningful relative genre rankings even when absolute probability calibration drops. This supports using Top-K prediction as a complementary strategy in practice.

Comparing our results with Unal et al. [1], who reported Hamming Losses around 9 to 10% on 29,432 images, the values observed here (19 to 21%) reflect the substantially smaller training set. Crucially, the relative ordering of architectures (ResNet50 > VGG16 > ConvNeXt) contradicts the original paper’s ranking, providing a novel insight: the optimal choice of transfer learning architecture is heavily dependent on the scale of the target dataset.

VI. CONCLUSION

This paper adapted the multilabel genre prediction framework of Unal et al. [1] to a constrained dataset of 300 IMDb movie poster images annotated with 14 genre labels. To mitigate overfitting, five pretrained CNN architectures were modified with a simplified linear classification head and evaluated using Hamming Loss as the primary metric, employing both a $\tau = 0.5$ threshold and Top-4 prediction strategies. ResNet50 achieved the best Hamming Loss of 0.1929 and the highest F1-Score of 0.6423 in the frozen backbone phase, outperforming VGG16 (0.2000), ConvNeXt (0.2014), MobileNet (0.2071), and DenseNet121 (0.2129). Full fine-tuning degraded performance for both top models, empirically confirming that frozen-feature transfer learning provides superior generalization under data scarcity.

Future work should investigate iterative stratification [8] for the train/test split to better preserve the genre distribution, explore label correlation modeling via graph-based methods, and consider multimodal fusion with plot summaries to improve prediction accuracy on rare genres such as Musical and Horror.

REFERENCES

- [1] F. Z. Unal, M. S. Guzel, E. Bostanci, K. Acici, and T. Asuroglu, “Multilabel genre prediction using deep-learning frameworks,” *Applied Sciences*, vol. 13, no. 15, p. 8665, 2023.
- [2] W.-T. Chu and H.-J. Guo, “Movie genre classification based on poster images with deep neural networks,” in *Proc. Workshop on Multimodal Understanding of Social, Affective and Subjective Attributes*, New York, NY, USA, 2017, pp. 39–45.
- [3] S. Sung and R. Chokshi, “Classification of movie posters to movie genres,” in *Proc. Workshop on Multimodal Understanding of Social, Affective and Subjective Attributes*, New York, NY, USA, 2017.
- [4] F. Zhuang, Z. Qi, K. Duan, D. Xi, Y. Zhu, H. Zhu, H. Xiong, and Q. He, “A comprehensive survey on transfer learning,” *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, 2021.
- [5] K. Kundalia, Y. Patel, and M. Shah, “Multi-label movie genre detection from a movie poster using knowledge transfer learning,” *Augmented Human Research*, vol. 5, p. 11, 2020.
- [6] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A ConvNet for the 2020s,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11966–11976.
- [7] M.-L. Zhang and Z.-H. Zhou, “A review on multi-label learning algorithms,” *IEEE Trans. Knowledge and Data Engineering*, vol. 26, no. 8, pp. 1819–1837, 2014.
- [8] K. Sechidis, G. Tsoumakas, and I. Vlahavas, “On the stratification of multi-label data,” in *Proc. Joint European Conf. on Machine Learning and Knowledge Discovery in Databases*, 2011, pp. 145–158.
- [9] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [10] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, 2016, pp. 770–778.
- [11] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, “Rethinking the inception architecture for computer vision,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, 2016, pp. 2818–2826.
- [12] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, 2017, pp. 4700–4708.
- [13] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, and H. Adam, “MobileNets: Efficient convolutional neural networks for mobile vision applications,” *arXiv preprint arXiv:1704.04861*, 2017.
- [14] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. Int. Conf. on Learning Representations (ICLR)*, San Diego, CA, 2015.